Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Development of HP-RTM-Based Mass Production Technology for Recycled-Carbon-Fiber Composite Marine Propellers*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding/marine system	*Main/auxiliary & engine system*	*Polymeric Materials*	*Composite Material Manufacturing Technologies*
Concept of Proposed Technology	The proposed technology focuses on the **development of composite marine propellers for small to medium-sized vessels** by applying **HP-RTM (High-Pressure Resin Transfer Molding)–based mass production processes**. The core concept is to improve productivity, cost competitiveness, and quality consistency of composite propeller blades, which have been limited to low-volume or prototype-level manufacturing. To achieve this, the project introduces a **sandwich-type composite structure**, in which **continuous carbon fibers are applied to the skin layers to satisfy strength and stiffness requirements**, while **recycled carbon fibers are used in the core layer**. This structural concept enables material cost			

	reduction and supports eco-friendly manufacturing without compromising propulsion performance. The proposed technology further emphasizes the **application of semi-automated preforming processes** suitable for HP-RTM, based on advanced manufacturing technologies developed by overseas research institutions. Through the integration of **preforming automation, HP-RTM mold design, and process parameter optimization**, the project aims to establish a manufacturing process capable of stable, repeatable, and high-quality propeller production. In addition, **simulation-based propeller design and performance analysis** are applied to ensure propulsion efficiency and structural reliability. The developed composite propellers will be validated through **real-vessel testing**, demonstrating their applicability to actual ship operations and supporting future commercialization in the shipbuilding and marine industry.
Rationale for Joint R&D Project	The rationale for this joint R&D project is based on the **need to secure core manufacturing technologies for the mass production and practical application of composite marine propellers**. While composite propellers offer advantages over conventional metal propellers in terms of weight reduction and performance improvement, domestic R&D activities have remained largely at the level of **design and performance analysis**, with **limited experience in mass production technologies and real-vessel validation**. In particular, the key technologies addressed in this project, including **HP-RTM-based composite propeller manufacturing and automated preforming processes**, are difficult to develop independently within a short period in Korea. The design of HP-RTM processes for **high-quality, repeatability, short cycle time**, including flow analysis, mold design, and process parameter optimization, as well as automated preforming technologies, represents **specialized technical domains accumulated over long-term research by leading overseas institutions**. (Development from the manual lay-up lamination to automated and stable HP-RTM technology for efficient upscaling strategy) Accordingly, this joint R&D project aims to enhance development efficiency and technical feasibility by combining the **domestic partner's capabilities in product design, performance evaluation, and real-vessel testing** with the **overseas research partner's expertise in automated preforming and core HP-RTM process technologies**. Through joint development and application of preforming process design and automation technologies possessed by the overseas partner, the project seeks to **establish a technically complete and reliable mass production process for composite marine propellers**. In addition, the application of **recycled carbon fibers in composite structural design and molding processes** represents an area with limited domestic experience in product-level implementation. Therefore, **international joint research is required** to evaluate material applicability, analyze molding characteristics, and optimize process conditions. Through this collaboration, the project aims to verify the industrial applicability of recycled carbon fibers and secure their effective integration into composite propeller manufacturing processes. Ultimately, this joint R&D project constitutes an **essential collaborative effort to internalize key manufacturing technologies that are difficult to secure through domestic efforts alone**, and to establish a commercialization foundation for composite marine propellers through **technical validation**

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	via real-vessel testing.
Key Words	1. RCF (Recycled Carbon Fiber) 2. HP-RTM (High Pressure Resin Transfer Molding) 3. Automated Preforming 4. Composite Propeller 5. Mass Production

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

The objective of this R&D project is to **secure core technologies for the mass-production application of composite marine propellers for small and medium-sized vessels**. To this end, the project covers technology development across the entire value chain, including **design, materials, manufacturing processes, and pre-commercial validation**.

1. **To secure structural and geometric design technologies for composite marine propellers.**
   This includes the development of outer mold line (OML) designs considering propulsion performance, as well as composite lay-up design to ensure structural reliability, thereby establishing composite propeller design technologies capable of replacing conventional metal propellers.
2. **To secure design technologies for sandwich-structured composite propellers using continuous and recycled carbon fibers.**
   Continuous carbon fibers are applied to the skin layers to ensure strength and stiffness, while recycled carbon fibers are utilized in the core layer to achieve both material cost reduction and environmental sustainability.
3. **To develop HP-RTM (High-Pressure Resin Transfer Molding)–based mass production process technologies for composite propellers.**
   This includes deriving HP-RTM process conditions capable of achieving short cycle times, uniform quality, and high dimensional accuracy, and establishing mass-production-suitable processes through mold design and optimization of process parameters.
4. **To develop semi-automated preforming process technologies compatible with HP-RTM.**
   Preforming processes enabling stable lay-up and shape realization of composite preforms containing recycled carbon fibers are applied to secure process repeatability and productivity.
5. **To secure performance and reliability validation technologies for composite propellers.**
   Simulation-based performance analysis is conducted in conjunction with experimental evaluation of manufactured prototypes, and the applicability of the developed propellers under actual operating conditions is verified through **real-vessel testing**.

R&D Target Specifications

The target performance objectives of this R&D project are defined based on key quantitative indicators to verify the feasibility of mass-production application of composite marine propellers. Quantitative targets are established to cover mechanical properties, manufacturing process performance, dimensional accuracy, and lightweight performance.

Evaluation Criteria	Unit	Proportion (%) of Each Item in Total	World's Highest Level (Country/Company) (Japan, Germany)	Domestic Level Before R&D	Development Goal Phase 1		Certification Criteria	Evaluation Method
			Performance level	Performance	Phase	Phase		

				level	1	2		
Tensile Strength (0/90)	MPa	50	700	700	650	700	ASTM D3039	Certified report
Flexural strength (0/90)	MPa	50	700	700	650	700	ASTM D3410	Certified report
HP-RTM process Time	min	20	Within 30 minutes	Within 30 minutes	≤27	≤25	-	Under the supervision of an inspector
Shape Deviation of Composite Propeller	mm	20	Class I	Class II	-	Class I	ISO-484-2	Certified report
Lightweight Composite Propeller Weight	kg	20	≤ 80	≤ 80	-	≤ 55	Certified test report scale	Under the supervision of an inspector

R&D Plan

Phase 1 (Basic Research Phase, 6 months)

- Lead Institution (MOTH Co., Ltd.):
  - Conduct performance investigations for targeted vessels.
  - Design the outer mold line (OML) for the primary propeller blade with a focus on propulsion performance.
  - Design the outer mold line (OML) for the secondary composite propeller blade with a focus on structural performance.
- International Partner (Fraunhofer ICT):
  - Investigate the feasibility of using rCF non-woven materials for the propeller core.
  - Conceptual design of molds for the prefabrication process and establishment of an overall process layout concept.
- Contracting Institution (Ulsan National Institute of Science and Technology):
  - Analyze the material properties influenced by the anisotropic structure of carbon fibers to facilitate the replacement of existing metal materials with CFRP materials, considering material selection, fiber stacking patterns, and molding processes.
  - Analyze the molding characteristics of recycled carbon fibers and establish a database of material properties.

Phase 2

- Lead Institution (MOTH Co., Ltd.):
  - Conduct real-ship testing: Perform propeller performance tests at sea.
  - Basic performance testing: Prepare in advance to acquire data during the real-ship tests, validating fundamental performance.
  - Analyze propulsion efficiency through real-ship tests of prototypes compared to existing products.
- International Partner (Fraunhofer ICT):
  - Based on Phase 1 test results, define the rCF core and carbon fiber layering through component design and simulation.
  - Conduct draping simulations, design molds for the prefabrication process, and manufacture prefabricated materials for HP-RTM propeller production.
  - Transfer technology related to fixtures for mass production and automated transport
  - LCA screening for the propeller and Database building
- Contracting Institution (Ulsan National Institute of Science and Technology):
  - Optimize process variables (pressure, temperature, injection speed, etc.) to produce initial prototypes.
  - Identify issues through trial production and implement improvements to create the final prototype.

- **Phased R&D Implementation Plan and SMART Milestones**

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	Detailed Task	Institution	Country	Phase 1		Phase 2 1st Year				Phase 2 2nd Year				Phase 2 3rd Year	
				Q1	Q2	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
WP1. Design of Ship Propellers Incorporating Recycled Carbon Fiber															
1-1	Performance Investigation of Target Vessels	MOTH	KOR												
1-2	Propulsion Performance Analysis	MOTH	KOR												
1-3	Propeller Design	MOTH	KOR												
1-4	Carbon Fiber Procurement	UNIST	KOR												
1-5	Composite Molding Using HP-RTM	UNIST	KOR												
1-6	Establishment of Continuous/Recycled Fiber Property Database	UNIST	KOR												
1-7	Composite Modeling and Structural Analysis	MOTH	KOR												
1-8	Design of DAQ for Vessel Performance Measurement	MOTH	KOR												
WP2. Development of Preform Automation Process															
2-1	Definition of Material Requirements and selection of Recycled Carbon Fiber	Fraunhofer ICT	GER												
2-2	Conceptual design of mold for preforming process and overall process layout concept	Fraunhofer ICT	GER												
2-3	Design and simulate process to define rCF core dimensions and carbon fiber layup based on testing results of Phase 1	Fraunhofer ICT	GER												
2-4	Draping simulation, Detailed preforming mold design and manufacturing of the preforms for propeller production with HP-RTM	Fraunhofer ICT	GER												
2-5	Concept development for jig system and high volume and efficiency production	Fraunhofer ICT UNIST	GER /KOR												
2-6	Technology transfer of the preform technology	Fraunhofer ICT MOTH	GER /KOR												
WP3. Prototype Production of Propellers Using HP-RTM															
3-1	Hub Design	MOTH	KOR												
3-2	Analysis of HP-RTM Parameters, Flow Analysis, and Mold Design	Fraunhofer ICT	GER												
3-3	Machining of Hubs and Components	MOTH	KOR												
3-4	Production of Trial Molds (Design/Manufacturing Technical Support)	MOTH (Fraunhofer ICT UNIST)	KOR /GER												
3-5	Analysis of Molding Characteristics for Sheet Unit Continuous/Recycled Carbon Fibers (Technical Support for HP-RTM Process Conditions)	UNIST (Fraunhofer ICT)	KOR												
3-6	Primary T/O of Propeller Blades Using HP-RTM (HP-RTM Process Technical Support)	UNIST (Fraunhofer ICT)	KOR /GER												
3-7	Identification of Issues in Trial Production and Process Improvements (Process Technical Consultation)	MOTH UNIST (Fraunhofer ICT)	KOR /GER												
3-8	Secondary T/O of Propeller Blades Using HP-RTM	UNIST	KOR												
3-9	Production of Final Improved Prototype	UNIST	KOR												
3-10	Design of Fixtures for Testing and Inspection	MOTH	KOR												
3-11	Production of Fixtures for Testing and Inspection	MOTH	KOR												
3-11	Non-Destructive Testing of Composite Propeller Blades	MOTH	KOR												
3-12	Balancing Testing of Composite Propellers	MOTH	KOR												
3-13	Production of Load Test Fixtures (Tensile, Bending, Fatigue)	MOTH	KOR												
3-14	Load Testing of Composite Propeller Blades	MOTH	KOR												

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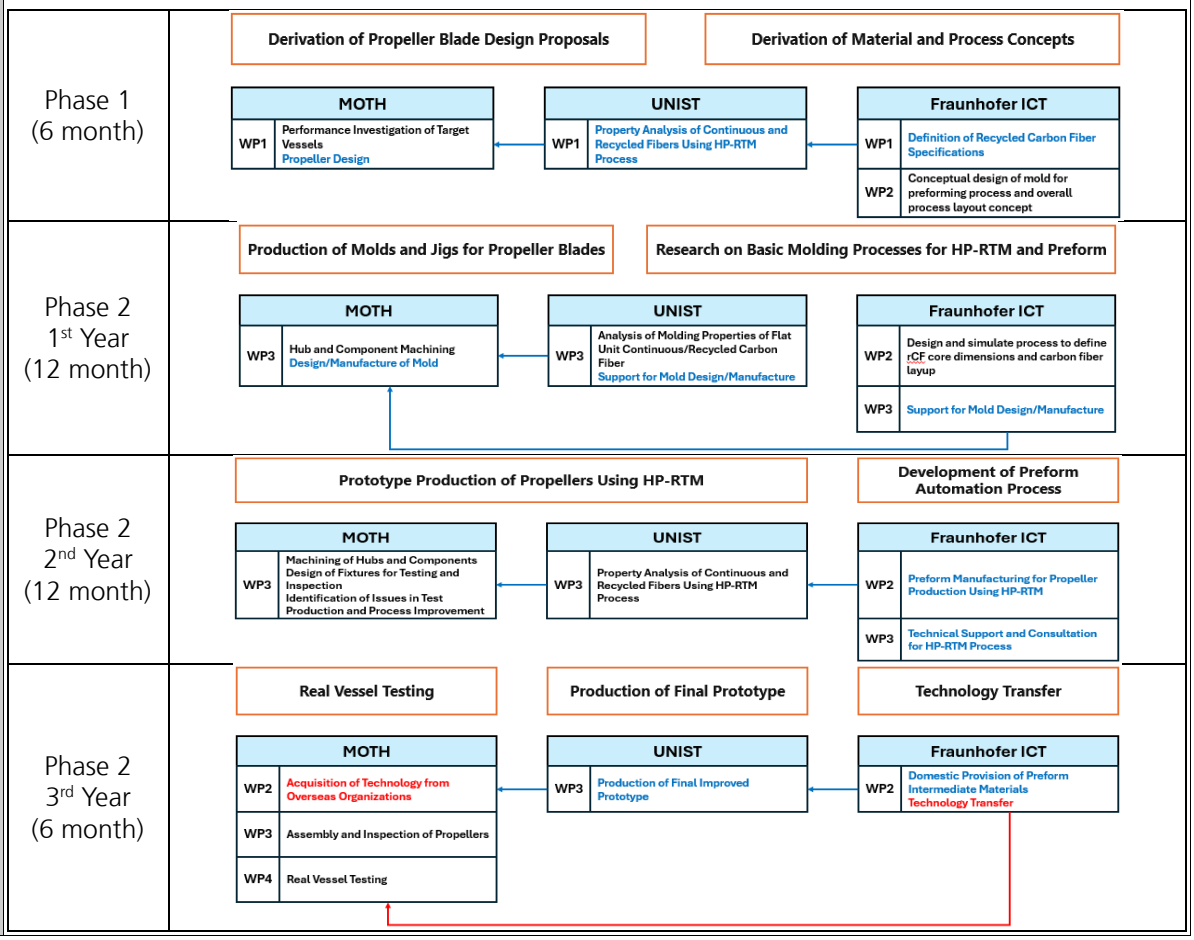
	(Tensile, Bending, Fatigue)		
3-15	LCA screening for the propeller and Database building	Fraunhofer ICT	GER
WP4. Real Vessel Testing			
4-1	Installation of DAQ on Target Vessels	MOTH	KOR
4-2	Real Vessel Testing	MOTH	KOR
4-3	Analysis of Vessel Propulsion Performance Data	MOTH	KOR
4-4	Analysis of Lift and Drag	MOTH	KOR

- **Detailed R&D Work Plan and Schedule by Work Package**



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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**International Research Institute: Fraunhofer ICT** • Fraunhofer ICT is expected to act as an overseas research institute providing advanced expertise in composite manufacturing processes and automation technologies, contributing to the joint development of core manufacturing technologies that are difficult to secure through domestic efforts alone. - HP-RTM process design and optimization support : Provision of expertise in process simulation, including resin flow analysis, and mold design concepts to enable high-quality and short-cycle HP-RTM molding. - Development of the preforming process based on automated concepts : Development of automated preforming concepts for continuous and recycled carbon fiber composites in marine sector. Further task with manufacturing the preform parts and process repeatability and stability assessment. - Analysis of molding defects and process improvement strategies : Technical analysis of trial molding results to identify defect mechanisms and recommend process improvement measures. - Technical consultation and knowledge exchange : Continuous technical consultation and joint research activities with domestic partners to facilitate effective knowledge sharing and technology internalization. • Through these contributions, Fraunhofer ICT is expected to function as a key technical partner supporting the joint development and maturation of advanced composite manufacturing and automation technologies. **Domestic Research Institute: UNIST** • UNIST is expected to serve as a domestic research institute responsible for experimental validation and process verification under local R&D environments, supporting the product-oriented development activities of the industrial lead organization. - Experimental verification of HP-RTM-based composite molding processes : Validation of HP-RTM process conditions at laboratory and pilot scales, including moldability evaluation and process repeatability verification. - Material characterization and database establishment for composite materials : Mechanical property evaluation and molding behavior analysis of continuous and recycled carbon fiber composites, and establishment of experimentally validated material property databases to support structural design. - Support for composite propeller prototype manufacturing and process validation : Technical support for preform lay-up, molding trials, and prototype fabrication to verify manufacturability and process stability. - Performance and reliability evaluation support : Analysis and interpretation of laboratory-scale test results to provide feedback for design and process improvement.. • Through these activities, UNIST is expected to play a key role in verifying the applicability and reliability of the developed technologies within the domestic R&D and manufacturing environment.
Expected R&R of Korean SME Partner	**Korean SME Lead Organization: MOTH** • MOTH is expected to act as the Korean SME lead organization, responsible for the overall execution and industrial application of this joint R&D project. The company takes primary responsibility for product development, system integration, testing, and preparation for commercialization of composite marine propeller technologies. - Overall project management and coordination : Planning and managing the project schedule, coordinating joint R&D activities, and maintaining communication among all participating organizations. - Definition of product requirements and system-level design : Defining target specifications and application requirements for composite marine propellers for small and medium-sized vessels.

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	- Design and development of composite marine propellers : Conducting propeller shape design and composite structural design, and updating designs based on test results. - Prototype manufacturing and integration : Producing composite propeller prototypes in cooperation with research institutes, including material preparation, molding coordination, and assembly. - Testing, validation, and real-vessel demonstration : Planning and conducting performance tests, durability tests, and real-vessel testing to verify applicability under actual operating conditions. - Provision of development resources : Providing engineering personnel, manufacturing facilities, and equipment required for prototype production and testing. - Collaboration and technology internalization : Actively collaborating with domestic and overseas research institutes to internalize key manufacturing technologies within the company. - Commercialization planning : Preparing plans for scale-up, mass production, and future market entry based on the project results. • Through these roles, MOTH Co., Ltd. is expected to lead the project from technology development to validation and commercialization readiness, ensuring effective industrial use of the R&D outcomes.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes** • The project is expected to deliver validated composite marine propeller technologies suitable for mass production and real-vessel application. Key technical outcomes include - Establishment of composite marine propeller design technologies capable of replacing conventional metal propellers for small and medium-sized vessels. - Development of stable and repeatable manufacturing processes suitable for industrial production of composite propellers. - Verification of structural performance, manufacturability, and reliability through prototype manufacturing and testing. - Demonstration of practical applicability under actual operating conditions through real-vessel testing. **Expected Industrial and Economic Impact** • The project is expected to contribute to strengthening the competitiveness of the Korean marine industry, particularly for small and medium-sized enterprises. - Reduction of dependency on imported metal propellers through domestic development of composite alternatives. - Improvement of productivity and cost competitiveness through mass-production-oriented manufacturing technologies. - Creation of new business opportunities for the Korean SME partner through product diversification and market expansion. - Establishment of a foundation for future commercialization and scale-up of composite marine propulsion components. **Expected Impact on Global Collaboration** • Through structured joint R&D with overseas research institutes, the project is expected to achieve: - Effective internalization of advanced manufacturing know-how through joint development rather than one-way technology transfer. - Strengthening of long-term international research collaboration frameworks.

	- Enhanced capability of Korean SMEs to participate in global value chains for composite marine components. **Expected Societal and Strategic Impact** • The project outcomes are expected to support broader national and industrial strategies by: - Contributing to lighter and more efficient marine propulsion systems, supporting improved operational efficiency. - Providing a technological basis applicable to future eco-friendly and advanced marine systems. - Supporting the transition of composite technologies from prototype-level research to practical industrial implementation.

6. References

(1) Shiyu Han, Peng Wang, Zhihui Jin, Xiaoyi An, Hui Xia, “Structural design of the composite blades for a marine ducted propeller based on a two-way fluid-structure interaction method,” Ocean Engineering, Volume 259, 1 September 2022.

(2) Shine Win Naung, Mahdi Erfanian Nakhchi, Mohammad Rahmati, “Numerical analysis of the performance of a composite marine propeller blade subject to structural blade oscillations,” Ocean Engineering, Volume 290, 15 December 2023.

(3) Firly Irhamni Ahsan, I Made Ariana, Aguk Zuhdi Muhammad Fathallah, "Blade Stress of Carbon Fiber B-Series Marine Propellers Based on Numerical Analysis," Journal of ETA Maritime Science 2024;12(3):277-286.

(4) Frank Henning, Luise Kärger, Dominik Dörr, Fabian J. Schirmaier, Julian Seuffert, Alexander Bernath, "Fast processing and continuous simulation of automotive structural composite components," Composites Science and Technology, Volume 171, 8 February 2019, Pages 261-279.

(5) Junhe Cui, Andrea La Spina, Jacob Fish, "Data-physics driven multiscale approach for high-pressure resin transfer molding (HP-RTM)," Computer Methods in Applied Mechanics and Engineering, Volume 417, Part A, 1 December 2023.

(6) Manseok Yoon, Minsu Ahn, "Study on molding control factors to reduce void contents in manufacturing CFRP parts by HP-RTM," Composites Part B: Engineering, Volume 296, 1 May 2025.

(7) Beom-Jeong Han, Yong-Chai Jeong, Sung-Ryul Kim, Ro-Won Kim, Myung Chang Kang, "Forming Characteristics with Cavity Pressure and Temperature Signal Inside Mold in High-Pressure Resin Transfer Molding Process of Carbon Fiber Reinforced Composite Material," Journal of the Korean Society of Manufacturing Process Engineers Vol.16 No.6 pp.81-86.

(8) Gwang-Chan Seong, Dong-Kyu Kim, Woong Han, Kwan-Woo Kim, Byung-Joo Kim, "Comparative analysis of carbon fiber reinforced composites: evaluating recycled carbon fibers as substitutes for commercial grades," Composites Part B: Engineering, Volume 307, 15 November 2025.

(9) So Young Oh, Chanwoo Joung, Seonghwan Lee, Yoon-Bo Shim, Dahun Lee, Gyu-Eun Cho, Juhyeong Jang, In Yong Lee, Young-Bin Park, "Condition-based maintenance of wind turbine structures: A state-of-the-art review," Renewable and Sustainable Energy Reviews, Volume 204, October 2024.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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